

Siqi Dai

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EDUCATION

University of Wisconsin–Madison

Bachelor of Science in Computer Science, Minor in Mathematics, GPA: 3.9/4.0

Madison, WI

Expected May 2028

TECHNICAL SKILLS

- **Programming:** Python, Java, C++, JavaScript, HTML/CSS, SQL
- **ML / AI:** PyTorch or TensorFlow, scikit-learn, Transformers, OpenCV, MediaPipe, XGBoost
- **Data / Scientific:** NumPy, Pandas, Matplotlib, signal processing

PROFESSIONAL EXPERIENCE

Summer of AI Laboratory Summer Intern & Team Leader

Jun 2026 – Aug 2026

SAIL

Madison, WI

- Led a four-member team developing SignBridge, a web-based ASL-to-speech system processing 543 hand, face, and pose landmarks per frame across a 2,044-sign-class recognition pipeline.
- Integrated temporal tracking, Transformer-based recognition, Gemini 2.5 language refinement, and text-to-speech, achieving 11.59 ms average local processing latency toward a 30 FPS target.

Full Stack Software Engineer Intern

May 2026 – Present

Rec App

Remote

- Developed mobile and backend features for a service-professional recommendation platform within an eight-developer Agile team.
- Implemented disaster-recovery procedures covering GitHub repository backups, Firestore backup verification, Firebase Authentication exports, and Firebase Storage inventory.

Student Research Assistant, AR/VR Accessibility Systems

Apr 2026 – Sep, 2026

Madability Lab, University of Wisconsin–Madison

Madison, WI

- Designed an experimental study protocol for evaluating AR stair-navigation assistance, specifying participant tasks, trial procedures, study conditions, and data-collection steps for future human-subject testing.
- Applied SAM 3 to segment ascending and descending stair environments, focusing on descending-stair navigation as an underexplored accessibility challenge.

Student Analyst

May 2026 – Present

Cybersecurity Operations Center

Madison, WI

- Investigate and triage 30+ phishing incidents weekly using WiscIT workflows across multiple university departments.
- Monitor Palo Alto Cortex XSOAR playbooks across network events for suspicious traffic, URL-filtering alerts, and potential threats.

PROJECT EXPERIENCE

Virtual Sensor-Guided Noise Diagnosis and Self-Correction Framework

May 2026 – Present

- Developed a residual-based framework integrating fault injection, 10-class sensor fault diagnosis, severity estimation, and signal correction.
- Achieved 98.74% classification accuracy and 98.56% macro F1 with XGBoost and reduced error by over 90% in representative bias and drift correction cases.

Multi-Parameter Virtual Sensing Using Neural Networks for Automotive Engines

Oct 2025 – Feb 2026

- Developed a multi-output ANN with a 6–64–32–16–3 architecture to estimate intake airflow, start of combustion, and engine-out NOx from six measured engine variables.
- Benchmarked the ANN against Random Forest, SVR, OLS, and linear-regression models and conducted permutation sensitivity analysis to rank sensor importance.

AWARDS & HONORS

- **Competition Awards:** American Society of Mechanical Engineers (ASME) Competition Winner; Central States Combustion Institute Undergraduate Research Competition Winner (2026)
- **Scholarships:** Mary Schreiber LeBahn & Duwayne LeBahn Scholarship; UW–Madison Undergraduate Scholarship for Summer Study (2026)
- **Academic Honors:** L&S Dean's List (2025, 2026)